order21 RESULTS
PREREG sha256 980ede89123837ceea8750b09c2c1303c1888cb5c117358b6389f6e847361f8b (hashed before numerics; Maps RG8a, RG8b hashed there).

test	value	verdict
K0 DFT reconstruction of C2b (x and −1/x, all)	max abs 2.1e−14	PASS
K1 reproduce order20 rel(X)	9.4949e−4 (diff 4e−18)	PASS
RG8a K=0: low / high	rel 2.7e−3 (abs 4.18) / rel 8.6e−3 (abs 4.33); joint rank 72	—
RG8a K=1: low / high	rel 3.2e−3 / 2.9e−2; joint rank 90	—
RG8a K=2: low / high	rel 3.4e−3 / 7.4e−2; joint rank 90	—
Map RG8a (low-|k| carries the near-law)	low rel ≥ 1e−3 for all K ≤ 2	**FAIL**
Map RG8b (smooth defect)	s(D) = 0.878 (control s(X) = 0.977)	**FAIL**
RG8b secondary (A2_corner defect)	s(D) = 0.855, s(X) = 0.977	(reported)

Reading
The order20 guess is falsified: truncating to low |k| makes the law worse (rel 2.7e−3 → 3.4e−3), not better than the full X (9.5e−4); the near-law is not localised in the k-spectrum. The defect is not smooth in x on [1/2,3], so there is no quantum-modular-type explanation of this shape.
Category A
Only the elementary fact $c_0 = B_2(\alpha_2)/N^2$ (multiplication theorem), so the k=0 term is $(B_2(\alpha_2) - 1/6)\cdot G_{\alpha,\beta}$. Nothing new warrants v8.
Category B
All table entries.

order21 NOTES
- Both new maps FAIL by preregistered rules; recorded before any post-hoc work. No post-hoc runs.
- Disclosed oddity (post-hoc observation, untested): every piece fits the w=1 law with κ within 1e−5 of 1 while the relative residual stays at 1e−3–1e−2. A κ that close to 1 with a non-vanishing residual suggests that a large exact component plus a small non-law component is present in each piece. One candidate: the order-t^0 factor x·kM in the weight-one expansion applied to a part of A2 that is P-even. Not tested.
- The k=0 piece equals diag(B2(α2)−1/6)·G; G obeys the x^{−1} law, so its approximate x^{+1} fit is a numerical coincidence of the 216-point grid until shown otherwise.
- $\bar{B}1\bar{B}1$ analogue of Prop. bern: proof is the 2D DFT version of order20 (i); not written to paper standard here; left for later.
- v8 not built (no Category A upgrade).
- Under the lock: no Π / ±1 grading / path predicate invented; MAPS_CANDIDATES/BATCH5_* untouched. E1–E3 OPEN. Ξ not identified.
- Run: nice, OMP_NUM_THREADS=1, <3 GB, ~16 min. The first foreground call was interrupted by the tool timeout; the process continued and completed normally (no rerun, no edits).